

TASK-2

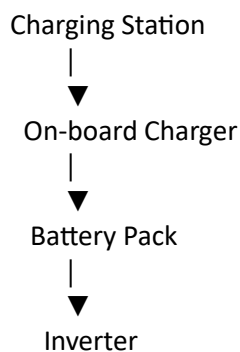
Question-1 Explain electric vehicle powertrain

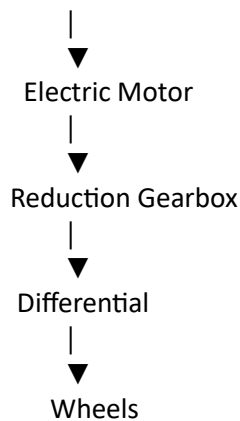
Solution:- An EV (Electric Vehicle) **powertrain** is the complete system that converts electrical energy stored in the battery into mechanical power to drive the wheels. Unlike a conventional vehicle, it does not use an internal combustion engine, fuel tank, or exhaust system. Instead, it relies on electric components to deliver power efficiently.

Main components of an EV powertrain

1. **Battery Pack**
 - Stores electrical energy.
 - Usually made of lithium-ion cells.
 - Determines the vehicle's driving range.
2. **Battery Management System (BMS)**
 - Monitors battery voltage, temperature, and state of charge.
 - Protects the battery from overcharging and overheating.
3. **Power Electronics**
 - **Inverter:** Converts DC (Direct Current) from the battery into AC (Alternating Current) for the motor.
 - **DC-DC Converter:** Steps down the high battery voltage to 12 V (or similar) for lights, infotainment, and other auxiliary systems.
 - **On-board Charger (OBC):** Converts AC power from a charging station into DC to charge the battery.
4. **Electric Motor**
 - Converts electrical energy into mechanical rotation.
 - Produces high torque from zero speed, giving EVs quick acceleration.
5. **Transmission (Reduction Gear)**
 - Most EVs use a single-speed reduction gearbox.
 - Reduces the motor's high rotational speed and increases torque to the wheels.
6. **Differential and Drive Axles**
 - Transfer power from the gearbox to the wheels.
 - Allow wheels to rotate at different speeds while turning.
7. **Thermal Management System**
 - Cools or heats the battery, motor, and electronics.
 - Maintains efficiency and extends component life

Working of an EV powertrain





During braking, the process can reverse:

Wheels → Motor → Inverter → Battery

This is called **regenerative braking**, where the motor acts as a generator to recover energy and recharge the battery

Question:2 Compare Brushless DC motor, Permanent magnet synchronous motor, Induction motor

Parameter	Brushless DC (BLDC) Motor	Permanent Magnet Synchronous Motor (PMSM)	Induction Motor (IM)
Operating Principle	Electronic commutation with trapezoidal back EMF	Sinusoidal AC supply with permanent magnets	Electromagnetic induction in rotor
Rotor	Permanent magnets	Permanent magnets	Conductive bars (squirrel cage), no magnets
Stator	Three-phase windings	Three-phase windings	Three-phase windings
Back EMF	Trapezoidal	Sinusoidal	Sinusoidal (induced)
Torque Output	Moderate torque ripple	Smooth torque	Smooth torque
Efficiency	High (85–95%)	Very high (90–97%)	High (85–94%)
Power Density	High	Very high	Moderate
Starting Torque	High	High	Moderate
Speed Range	Wide	Wide	Very wide
Control Method	Six-step commutation	Field-Oriented Control (FOC)	Vector Control (FOC)
Maintenance	Very low	Very low	Very low
Cost	Medium	High (rare-earth magnets)	Lower (no magnets)
Reliability	High	High	Very high
Heat Generation	Mainly stator	Mainly stator	Rotor and stator
Regenerative Braking	Supported	Excellent	Supported

Parameter	Brushless DC (BLDC) Motor	Permanent Magnet Synchronous Motor (PMSM)	Induction Motor (IM)
Noise & Vibration	Higher due to torque ripple	Very low	Moderate

Question:3 Explain Battery management system in detail

A Battery Management System (BMS) is an electronic control system that monitors, protects, and manages the rechargeable battery pack in an electric vehicle (EV). It ensures the battery operates safely, efficiently, and reliably while maximizing its performance and lifespan.

Requirement of battery management system

An EV battery pack consists of hundreds or thousands of lithium-ion cells connected in series and parallel. These cells are sensitive to:

- Overcharging
- Over-discharging
- High current
- High temperature
- Cell imbalance

Without a BMS, these conditions can reduce battery life, degrade performance, or even lead to thermal runaway and fire.

Functions of a Battery Management System

Battery Management System (BMS) monitors, controls, and protects the battery pack in an electric vehicle. Its main functions are:

1. Battery Monitoring

- Continuously measures the voltage of each cell.
- Monitors battery current (charging and discharging).
- Measures the temperature of cells and the battery pack.
- Detects abnormal operating conditions.

2. State of Charge (SOC) Estimation

- Calculates the remaining battery capacity.
- Displays the battery percentage to the driver.
- Helps estimate the remaining driving range.

3. State of Health (SOH) Estimation

- Determines the overall condition of the battery.
- Monitors battery aging and capacity loss.
- Predicts the remaining useful life of the battery.

4. Cell Balancing

- Ensures all cells have nearly equal voltage.
- Prevents overcharging and over-discharging of individual cells.
- Improves battery performance and extends battery life.

5. Battery Protection

Protects the battery against:

- Over-voltage
- Under-voltage
- Over-current

- Short circuit
- Over-temperature
- Under-temperature
- Reverse polarity

If unsafe conditions occur, the BMS disconnects the battery using contactors or MOSFETs.

6. Thermal Management

- Monitors battery temperature.
- Controls cooling and heating systems.
- Maintains the battery within the optimum operating temperature range.

7. Charging Control

- Regulates charging voltage and current.
- Prevents overcharging.
- Stops charging when the battery reaches full charge.
- Supports safe fast charging.

8. Power Management

- Limits battery power when the battery is low or too hot.
- Controls charging and discharging rates.
- Optimizes energy usage.

9. Communication

- Communicates with the vehicle control unit (VCU), motor controller, charger, and dashboard.
- Uses communication protocols such as **CAN Bus** to exchange battery status and fault information.

CASE STUDY

TATA NEXON EV

Case Study: Tata Nexon EV (10 Marks)

Introduction

The **Tata Nexon EV** is an electric compact SUV manufactured by [Tata Motors](https://www.tatamotors.com). Introduced in **2020**, it has become one of India's most popular electric vehicles due to its performance, safety, affordability, and low running cost.

Technical Specifications

- **Motor:** Permanent Magnet Synchronous Motor (PMSM)
- **Battery:** Lithium-ion battery (30 kWh or 45 kWh, depending on variant)
- **Drive Type:** Front-Wheel Drive (FWD)
- **Transmission:** Single-speed automatic
- **Top Speed:** Approximately 150 km/h
- **Charging:** AC charging and DC fast charging

Key Features

- High-efficiency PMSM motor with instant torque
- Advanced Battery Management System (BMS) for battery safety and performance
- Regenerative braking to recover energy during braking
- Liquid-cooled battery for better thermal management
- Fast charging capability
- Zero tailpipe emissions and low maintenance

Advantages

- Environment-friendly with zero emissions
- Low running and maintenance costs
- Smooth and quiet driving experience
- Good driving range for city and highway use

- Modern safety features and reliable performance

Limitations

- Higher initial purchase cost than petrol/diesel vehicles
- Charging time is longer than conventional refueling
- Driving range may reduce under heavy loads or extreme weather
- Charging infrastructure is still developing in some areas

Conclusion

The **Tata Nexon EV** is a successful example of modern electric vehicle technology in India. Its efficient PMSM motor, lithium-ion battery, Battery Management System (BMS), and regenerative braking provide excellent performance, safety, and energy efficiency. It has played a significant role in promoting electric mobility and sustainable transportation in the country.

REFERERNCES USED

- PDF provided by maincraft technology
- Youtube
- Google gemini